

Arman Phaugat

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Summary

Backend-focused developer who improved RAG pipeline latency by 3.5x (28s to 8s) and cut database query time by 65% while building production-grade systems in Node.js, Python, and FastAPI. Skilled in designing RESTful APIs, SQL/NoSQL databases, caching (Redis), and message queues (BullMQ), with hands-on experience in JWT authentication, rate limiting, Docker, and AI/RAG pipeline integration.

Education

Bachelor of Technology in Computer Science and Engineering

Manipal University Jaipur

Jaipur, Rajasthan

August 2023 – July 2027

Current CGPA: **9.05**

Experience

Web Development Intern

Indavis Lifesciences, Haridwar

June 2025 – July 2025

- Maintained and updated **7+ web pages/routes**, ensuring smooth performance and content accuracy.
- Collaborated with teams to align website updates with **brand and business objectives**.

Projects

Nori - Founder & Backend/AI Dev (Live | GitHub)

FastAPI, PostgreSQL, Redis, BullMQ, Discord.py, Groq, Graphlit, Tavily, Podman, Oracle Vps, Caddy

- AI chatbot platform for organizations – lets teams spin up a custom chatbot in under 10 minutes with support for 10+ file type ingestions.
- Improved RAG pipeline latency from **28s to 8s** (3.5x improvement) through profiling and eliminating bottlenecks in the retrieval-augmented generation process.
- Optimized retrieval throughput by replacing sequential database queries with **parallel requests**, reducing per-query database time by **65%**.

ZymeRag – Open-Source Multimodal RAG Platform (GitHub)

FastAPI, PostgreSQL, Redis, BullMQ, FAISS, BM25, Groq, Docker

- Built ZymeRag as the **RAG engine behind Nori**, our startup, and used it in the **Deloitte Capstone** open-sourced it after Graphlit, the managed platform we relied on, shut down.
- Designed **hybrid retrieval** (BM25 keyword search + FAISS semantic search) with a **separate index per document**, running both searches in **parallel** across selected documents to return relevant chunks in **~3 seconds**.
- Automated **24-hour chunk updates** for saved websites using **BullMQ/Redis** jobs with **batching**.

Real Time Stock Trading Backend (Demo | GitHub)

Node.js, Express, MySQL, Redis, BullMQ, JWT, Docker, Argon2, Postman

- Built high-performance leaderboard system using Redis Sorted Sets with pipeline optimization, **achieving <50ms query time**.
- Engineered automated price simulation engine using **BullMQ + Redis with 30-minute repeatable jobs**, updating stock prices with $\pm 25\%$.
- Secured APIs with JWT authentication (access + refresh tokens), custom Token Bucket **rate limiter at 5 req/sec per user**, and containerized Redis using **Docker for BullMQ job** processing and leaderboard cache management.

Technical Skills

Programming: Python, JavaScript, C, C++

Web Technologies: Node.js, Express.js, FastAPI, REST APIs

Database: MySQL, MongoDB, Redis, FAISS, SQLite3

Libraries/Frameworks: LangChain, HuggingFace, Discord.py, JWT, Bcrypt, Argon2, Matplotlib, Express.js, BullMQ

Tools & Platforms: Docker, Git, GitHub, VS Code, Postman API, Cloudinary, Groq

Core Concepts: Data Structures, Algorithms, System Design, OOP, Operating Systems, Machine Learning, ACID Transactions, Caching, Rate Limiting, Authentication & Authorization, Vector Databases, RAG, RESTful API Design, Microservices, Event-driven Architecture, CI/CD

Achievements

- Received Dean's Excellence Award across multiple semesters for maintaining a 9.0+ CGPA.
- Solved **1200+ DSA problems** across various platforms.
- LeetCode: Ranked in the top **0.3%** globally in terms of Problem Solved.
- Ranked **13th** in C-Score (DSA, Development, CP & Consistency) and **10th** all-time for total problems solved, across MIT & MUJ on Codolio.
- Selected for Deloitte Capstone Team, ranked in the **top 20** among 400+ competing teams.
- Advanced to the **Semi-Final Round of MUJHackX Hackathon** among 1300+ participants at Manipal University Jaipur.
- Advanced to the **Semi-Final Round of ZS Campus Beats 2026** among **7,000+ candidates** and **2,450+ teams**.